

OTTAWA INTL, ON, Canada

WMO: 716280

Lat: 45.3171N

Lon: 75.6739W

Elev: 115

StdP: 99.95

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.1	-21.4	-30.4	0.2	-23.3	-27.6	0.3	-20.9	12.0	-6.5	10.8	-8.8	3.7	290	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.4	30.9	22.0	29.2	20.9	27.6	20.2	23.5	28.6	22.4	27.0	21.5	25.8	4.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.9	16.8	26.2	20.9	15.8	24.9	20.0	14.9	24.0	70.7	28.7	66.6	27.2	63.1	25.7	
(p)															

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	9.0	8.1	-27.0	33.6	2.9	1.8	-29.1	34.9	-30.8	35.9	-32.5	36.9	-34.6	38.2		
(5)			-27.3	25.4	2.9	1.2	-29.4	26.3	-31.1	27.0	-32.7	27.6	-34.8	28.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	6.8	-9.8	-7.9	-2.1	6.0	13.6	18.8	21.2	20.2	15.9	8.7	1.7	-5.3
	DBStd	11.90	7.15	6.08	5.94	4.90	4.42	3.77	2.88	2.96	4.09	4.50	5.13	6.22
	HDD10.0	2448	613	500	375	137	15	0	0	0	3	79	252	474
	HDD18.3	4451	871	733	632	369	157	39	7	16	96	299	499	733
	CDD10.0	1292	0	0	1	18	128	263	347	314	178	40	2	0
	CDD18.3	254	0	0	0	1	11	53	96	72	22	1	0	0
	CDH23.3	1998	0	0	1	11	128	436	748	501	168	6	0	0
	CDH26.7	536	0	0	0	2	30	122	217	127	38	0	0	0
Wind	WSAvg	3.8	4.2	4.2	4.3	4.4	3.8	3.4	3.2	3.1	3.2	3.8	4.0	4.1
Precipitation	PrecAvg	939	70	54	62	77	76	94	90	85	92	86	74	80
	PrecMax	1182	124	104	127	156	127	192	196	163	160	182	111	136
	PrecMin	793	30	23	23	14	27	17	18	38	44	22	23	26
	PrecStd	93	25	21	28	34	27	39	42	24	33	34	24	29
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.6	7.2	17.1	24.9	29.7	32.3	33.3	32.4	30.3	24.1	16.8	10.6
		MCWB	7.9	4.9	12.5	16.2	20.2	22.8	23.7	22.4	21.7	17.6	12.8	9.0
	2%	DB	4.7	4.5	11.6	20.3	26.8	29.9	30.8	29.7	27.4	20.8	14.1	7.0
		MCWB	3.5	2.4	7.4	12.4	17.9	21.3	22.2	21.3	20.7	15.8	11.6	5.8
	5%	DB	2.4	2.7	8.2	16.8	24.1	27.8	29.1	27.8	25.1	18.1	11.4	4.2
		MCWB	1.3	1.0	4.8	10.6	16.1	20.0	21.2	20.5	19.2	14.1	8.7	3.0
	10%	DB	0.4	1.1	5.6	14.1	21.7	25.8	27.4	26.2	23.0	15.7	8.9	2.5
		MCWB	-0.4	-0.2	2.7	8.4	14.9	19.0	20.4	19.8	18.2	12.4	6.5	1.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.2	5.6	12.8	17.2	21.9	24.3	25.5	24.2	23.3	19.1	14.1	9.3
		MCDB	8.5	6.6	16.6	22.9	27.7	30.4	31.4	30.0	27.6	22.3	15.8	10.2
	2%	WB	3.7	2.7	7.9	14.0	19.5	22.7	23.8	22.7	21.7	16.5	11.8	5.8
		MCDB	4.4	4.1	11.0	18.0	24.7	27.8	28.6	27.3	25.8	19.8	14.0	7.0
	5%	WB	1.4	1.2	5.0	11.2	17.7	21.4	22.6	21.8	20.2	14.6	9.1	3.3
		MCDB	2.3	2.4	7.5	15.4	21.9	25.9	27.1	25.9	24.0	17.5	11.0	4.2
	10%	WB	-0.3	-0.1	3.1	9.1	16.0	20.1	21.5	20.7	18.9	12.9	6.9	1.4
		MCDB	0.5	1.0	5.4	13.5	20.3	24.1	25.8	24.4	22.2	15.3	8.6	2.3
Mean Daily Temperature Range	5% DB	MDBR	8.2	8.6	8.6	9.9	10.8	10.3	10.4	10.3	10.3	8.5	7.4	6.9
		MCDBR	9.8	9.1	11.5	15.2	14.7	13.1	12.7	12.4	12.8	12.2	11.0	8.0
		MCWBR	9.5	8.2	8.2	9.2	7.8	6.5	6.1	5.8	6.9	7.8	8.9	7.5
	5% WB	MCDBR	9.5	8.8	11.0	13.2	12.2	11.1	11.0	10.5	11.5	11.0	10.5	8.1
		MCWBR	9.4	8.3	8.7	9.3	7.7	6.3	6.1	5.7	7.0	7.8	9.1	7.7
Clear-Sky Solar Irradiance	taub	0.263	0.287	0.322	0.344	0.380	0.413	0.417	0.400	0.368	0.338	0.308	0.278	
	taud	2.385	2.312	2.362	2.416	2.353	2.273	2.264	2.320	2.400	2.481	2.497	2.403	
	Ebn at Noon	860	897	910	917	888	856	847	849	849	830	800	798	
	Edh at Noon	72	95	105	109	120	131	131	119	101	81	67	65	
All-Sky Solar Radiation	RadAvg	1.39	2.35	3.67	4.48	5.20	5.65	5.75	5.06	3.97	2.34	1.43	1.02	
	RadStd	0.16	0.23	0.31	0.46	0.51	0.44	0.48	0.24	0.33	0.21	0.16	0.09	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page